A CAPSTONE PROJECT – I REPORT

ON

"PRAGATI AI "

Submitted in partial fulfillment of the requirements

for the award of the degree of

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IN

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CERTIFICATE

This is to certify that the Capstone Project - I Report entitled "PRAGATI AI" submitted by TIRTH PATEL & RANJAN KUMAR Enrollment No: 2301201132 & 2301201143 in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering/ Computer Engineering has been successfully completed during the academic year 2026-2027. This work has been carried out under my guidance and supervision and it is up to my satisfaction.

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DECLARATION

We hereby declare that the Capstone Project – I Report entitled **"PRAGATI AI "** is a record of the original work carried out by me during the course of my Capstone Project – I. This work has not been submitted, either in part or in full, to any other University or Institution for the award of any degree or any other academic qualification.

I further declare that all sources of information used in this report have been duly acknowledged and cited wherever necessary.

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ABSTRACT

PRAGATI AI is an AI-powered interactive learning assistant designed to enhance the traditional learning environment through intelligent automation and real-time interaction. The system provides students and teachers with an interactive platform where they can ask academic questions and receive instant AI-generated responses. It aims to make learning more accessible, engaging, and efficient.

The system includes secure signup and login functionality for managing user accounts and storing user information. After authentication, users can access the chatbot interface to ask questions and obtain relevant solutions through an integrated AI API. The platform supports personalized learning, quick doubt resolution, and continuous learning beyond traditional classroom hours.

The project is developed using technologies such as HTML, CSS, JavaScript, PHP, and MySQL, along with AI-based tools. By integrating these technologies, PRAGATI AI helps reduce the dependency on manual assistance, improves communication between students and teachers, and provides a user-friendly digital learning environment.

Overall, PRAGATI AI aims to bridge the gap between traditional classroom learning and modern AI-powered education by providing instant support, interactive learning, and an efficient platform for both students and teachers.

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CHAPTER 1: OVERVIEW OF THE PROJECT

1.1 Project Overview

PRAGATI AI is designed to create an intelligent and interactive learning platform that enhances the learning experience through the integration of Artificial Intelligence (AI) technology. The system allows both students and teachers to sign up and log in securely, ensuring that their personal and academic data is safely stored and managed in the database.

Once logged in, users can access an interactive chatbot interface where they can ask academic questions and receive AI-generated responses in real time. PRAGATI AI currently supports conversations in three languages: English, Hindi, and Gujarati, allowing users to communicate with the chatbot in their preferred language.

The platform also provides a file upload feature through which users can upload files and interact with the information or content provided through the chatbot. In addition to text-based interaction, PRAGATI AI includes a microphone option that allows users to speak their questions instead of typing them. The chatbot can listen to the user's voice input, process the question, and provide an appropriate response, making the interaction more convenient and accessible.

The system uses an integrated API to process user queries and generate AI-powered responses. It combines secure authentication, multilingual conversation, file uploading, text-based interaction, and voice-based interaction in a single learning platform. The project is developed using technologies such as HTML, CSS, JavaScript, PHP, MySQL, and XAMPP.

Overall, PRAGATI AI aims to bridge the gap between traditional learning methods and modern AI-powered education by providing students and teachers with instant academic assistance, multilingual communication, file-based interaction, and voice-enabled chatbot functionality.

1.2 Problem Statement

Traditional classroom learning often depends on fixed teaching methods and scheduled learning hours, which can make it difficult for students to receive immediate support when they have doubts. Students may also have limited access to interactive learning tools and personalized assistance outside the classroom. Teachers can face difficulties in providing individual attention to every student while also managing regular academic activities.

Another challenge is the limited availability of real-time and flexible communication between users and learning systems. A conventional learning environment may not provide an intelligent platform where students and teachers can ask academic questions and receive instant responses according to their requirements.

To address these challenges, PRAGATI AI provides an AI-powered interactive learning platform that allows students and teachers to communicate with an intelligent chatbot and obtain real-time academic assistance. The system also supports multilingual interaction in English, Hindi, and Gujarati, file uploading for interacting with uploaded content, and voice-based interaction through a microphone. These features make the learning process more interactive, accessible, and convenient.

The main problem addressed by PRAGATI AI is the gap between traditional classroom learning and the need for instant, personalized, interactive, and accessible academic assistance. The system aims to reduce dependency on physical assistance and provide learning support beyond traditional classroom hours. The original AIGNUS report identifies delayed doubt resolution, limited interactive learning tools, and the need for real-time support as key problems motivating the system

1.3 Objectives

Table 1.1: Project Objectives

S. No.	Objective	Description
1	Interactive Learning Platform	To provide a smart and interactive learning platform that allows students and teachers to access instant solutions to academic queries.
2	Secure User Authentication	To implement secure signup and login functionality that ensures safe storage and management of user data.
3	Real-Time Chatbot	To enable real-time communication through a chatbot interface, allowing users to ask questions and receive accurate responses powered by an API.
4	AI-Powered Problem Solving	To simplify classroom learning by integrating AI technology for quick and reliable problem-solving.
5	24/7 Learning Support	To reduce the dependency on physical assistance by offering a 24/7 learning support system.
6	User Engagement	To create a user-friendly environment that enhances engagement and participation in the learning process.
7	Knowledge Retention	To provide an interactive learning environment that supports better knowledge retention.
8	Accessible Learning	To make education more accessible and efficient through the integration of AI-powered learning assistance.
9	Traditional and Digital Learning	To bridge the gap between traditional and digital learning by providing an intelligent and interactive educational platform.

1.4 Scope of the Project

The PRAGATI AI system is designed to provide an interactive and intelligent learning platform that connects students and teachers through an AI-powered assistant. The system provides secure signup and login functionality so that users can access the platform using their credentials. After signing in, students can access the chat page to ask academic questions and receive AI-generated responses instantly. Teachers can also use the chatbot to ask academic or subject-related questions and receive AI-powered assistance for lesson preparation, doubt clarification, and classroom activities. The platform aims to provide learning support anytime and encourage self-paced learning.

PRAGATI AI also extends the learning experience through multilingual communication, allowing users to interact with the chatbot in English, Hindi, and Gujarati. The system includes a file upload facility that allows users to upload files and interact with the information contained in those files. In addition to text-based communication, the platform provides a microphone option through which users can speak their questions. The chatbot can listen to the spoken question, process it, and provide an appropriate response, making interaction more convenient and accessible.

The overall scope of PRAGATI AI is to provide a centralized AI-based learning assistant for both students and teachers. It combines secure access control, personalized interaction, AI-powered question answering, multilingual communication, file-based interaction, and voice-enabled interaction within a single platform. The system is intended to improve engagement, accessibility, and efficiency in the learning process while reducing dependency on physical assistance. The platform can also be extended in the future with advanced analytics, classroom activity tracking, progress monitoring, and integration with learning management systems.

1.5 Organization Chart

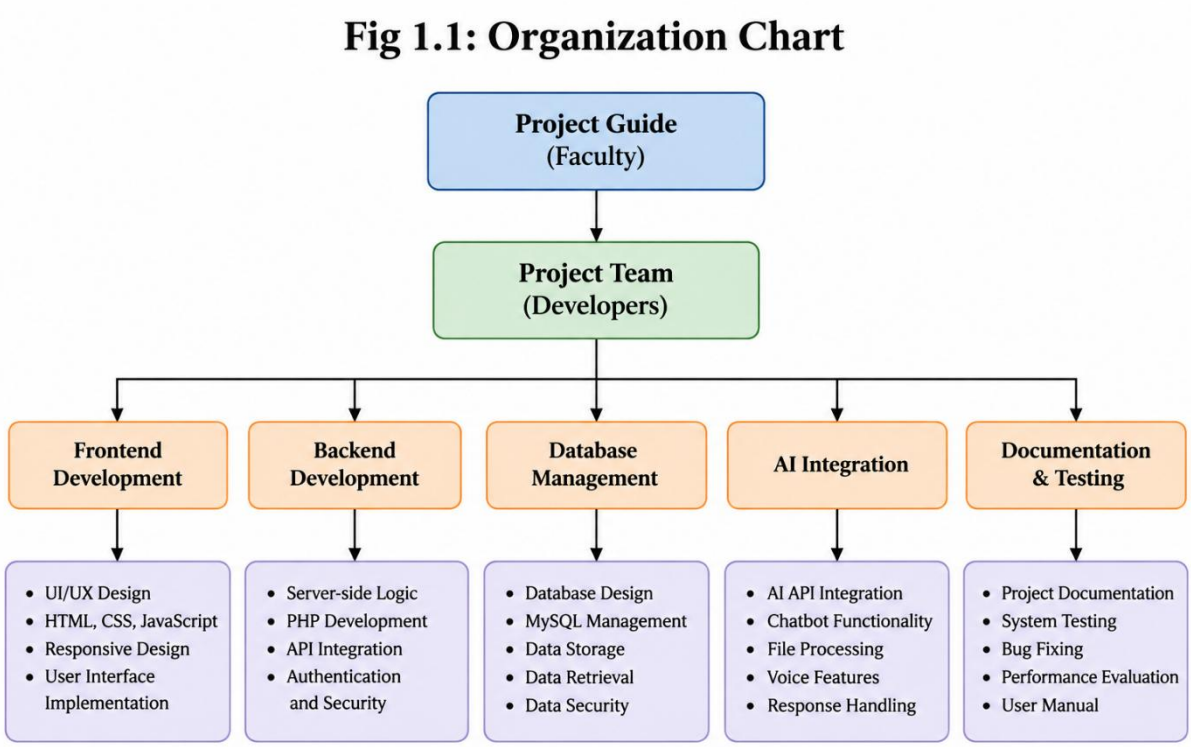


Figure 1.1

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

The development of Artificial Intelligence has created new opportunities for improving the education sector. Traditional classroom learning generally depends on fixed teaching methods, scheduled learning hours, and direct support from teachers. These limitations can make it difficult for students to receive immediate assistance when they have doubts or need additional learning support. AI-based educational systems provide a way to overcome these limitations by offering automated assistance, personalized learning support, and real-time interaction. Such systems can improve student engagement and make the learning process more accessible and efficient.

AI technology can also help create smarter classroom environments by supporting both students and teachers. Students can receive instant assistance for their academic queries, while teachers can use AI-based tools to obtain information and support lesson preparation and classroom activities. This reduces dependency on manual assistance and allows learning support to be available beyond regular classroom hours. The use of AI therefore provides an opportunity to combine traditional educational methods with modern digital technologies.

2.2 Review of Existing Systems

Existing AI-based learning systems are developed to provide students and teachers with faster and more convenient access to academic assistance. Traditional learning environments generally depend on classroom hours and direct support from teachers, which may delay doubt resolution and reduce student engagement. AI-based systems address this limitation by providing automated assistance and real-time interaction through intelligent chatbot interfaces.

The PRAGATI system provides a basic AI-powered learning environment with secure signup and login functionality for students and teachers. After authentication, users can access a chatbot interface to ask academic questions. The system sends user queries through an integrated API and receives AI-generated responses in real time. User information and chat history are maintained using a MySQL database, allowing the system to provide organized and secure access to the platform.

The existing system provides useful functionality for academic assistance, but it has certain limitations. It mainly focuses on basic authentication and chatbot interaction and does not provide advanced performance analytics, progress tracking, or administrative control. Personalization is also limited, and the user experience can be affected by internet connectivity and API availability. These limitations indicate the need for an enhanced learning platform with more flexible and interactive methods of communication.

PRAGATI AI addresses these limitations by extending the basic chatbot approach with multilingual communication in English, Hindi, and Gujarati, file upload and interaction, microphone-based question input, and chatbot voice-listening functionality. These additional capabilities provide users with more flexible ways to interact with the AI assistant and make the learning platform more accessible and interactive.

2.3 Proposed System

PRAGATI AI is an enhanced AI-powered learning platform designed to provide students and teachers with intelligent and interactive academic assistance. The system provides secure user authentication and real-time chatbot-based responses for academic queries. It supports communication in English, Hindi, and Gujarati, allowing users to interact with the chatbot in their preferred language. Users can also upload files and interact with the information contained in the uploaded content. The system includes a microphone option that allows users to speak their questions instead of typing them, along with a voice-listening mode through which the chatbot can hear and process the user's questions. The platform is developed using HTML, CSS, JavaScript, PHP, and MySQL and provides a simple and user-friendly learning environment. Overall, PRAGATI AI aims to make learning more accessible, interactive, flexible, and efficient through the integration of AI technology.

CHAPTER 3: TOOLS & TECHNOLOGIES USED

3.1 Software Used

Table 3.1: Software Used

S. No.	Software / Technology	Purpose / Usage
1	HTML	Used to create the structure and content of the web pages of the PRAGATI AI system.
2	CSS	Used to design and style the user interface and improve the visual appearance of the web pages.
3	JavaScript	Used to provide interactive functionality and handle user-side actions in the application.
4	PHP	Used for backend processing, server-side functionality, user authentication, and communication with the database.
5	MYSQL	Used as the database management system for storing user information and chat-related data.
6	XAMPP	Used as the local server environment for running the PHP application and MySQL database.
7	Git / GitHub	Used for managing and maintaining the project source code and version control.
8	Postman	Used for testing and checking API requests and responses during system development.
9	Visual studio code	Used as the development environment for writing, editing, and managing the project source code.

The PRAGATI AI system is developed using a combination of software tools and technologies to support its frontend, backend, database, and API-related functionality. HTML is used to create the basic structure and content of the web pages and provides the foundation of the application interface. CSS is used to design the webpages and improve their overall appearance, layout, and presentation. JavaScript is used to provide interactive functionality and handle different client-side activities. PHP is used for backend processing and server-side operations, including communication between the application and database. MySQL is used as the relational database management system for storing user information and chat-related data. XAMPP provides a local server environment required to run the PHP application and MySQL database during development. Git and GitHub are used for managing and maintaining the project source code and supporting version control. Postman is used for testing API requests and responses during development and integration. Visual Studio Code is used as the development environment for writing, editing, and managing the project source code. These software technologies work together to provide a functional, interactive, and organized environment for developing PRAGATI AI. The combination of frontend technologies, backend processing, database management, and development tools helps in building a reliable and user-friendly learning platform.

3.2 Hardware Used

Table 3.2: Hardware Used

S. No.	Hardware Component	Specification Requirement	Purpose / Usage
1	Computer / Laptop	Intel Core i3 equivalent processor or above	Used for developing, running, and testing the PRAGATI AI web application.
2	RAM	Minimum 4 GB, 8 GB recommended	Supports the development environment, browser, server, and database operations.
3	Storage	Minimum 20 GB available Disk space	Used for storing project files, software, database, and dependencies.
4	Display	Minimum 1280 × 720 resolution	Used for viewing and testing the application interface properly.
5	Keyboard	Standard keyboard	Used for coding, entering data, and interacting with the application
6	Mouse	Standard pointing device	Used for navigating the development environment and web application.
7	Internet Connection	Stable broadband / internet connection	Required for API communication and accessing online services used by the application.

Note: The system does not require specialized hardware. A standard computer or laptop with the above specifications is sufficient for developing and running the PRAGATI AI system.

3.3 Programming Languages / Technologies

PRAGATI AI is developed using a combination of programming languages and web technologies that work together to provide an interactive and intelligent learning platform. The frontend of the system is developed using HTML, CSS, and JavaScript, while PHP is used for backend processing and MySQL is used for database management. XAMPP provides the local development environment required for running and testing the application. The system also uses an AI API to process user questions and provide AI-generated responses in real time.

HTML, or HyperText Markup Language, is used to create the basic structure of the PRAGATI AI web pages. It defines important elements such as headings, paragraphs, forms, buttons, input fields, navigation sections, and other components required for the user interface. HTML provides the structural foundation of the application and is used for developing pages such as the home page, signup page, login page, and chat page. A proper HTML structure helps organize the content and allows other technologies such as CSS and JavaScript to be applied effectively.

CSS, or Cascading Style Sheets, is used to control the visual presentation of the PRAGATI AI application. It is responsible for the layout, spacing, alignment, dimensions, fonts, buttons, forms, and other visual elements of the interface. CSS helps maintain a consistent design throughout the application and improves the readability and usability of the webpages. It also helps create a clean and responsive interface that can be used across different supported devices. The use of CSS separates the visual design from the basic HTML structure, making the interface easier to modify and maintain.

JavaScript is used to provide interactive and dynamic functionality within the PRAGATI AI system. It allows the application to respond to user actions and perform specific operations within the browser without requiring every interaction to reload the entire webpage. JavaScript can be used to handle interface events, manage interactive elements, and provide smoother navigation and communication within the application. It contributes to the responsive nature of the chat interface and improves the overall interaction between the user and the system.

PHP is used as the server-side scripting language for the backend of PRAGATI AI. It manages the flow of data between the frontend interface, database, and AI service.

PHP handles important operations such as user authentication, processing requests, managing sessions, communicating with the database, and handling responses from the AI API. The backend provides the required processing layer between the user's actions and the system's stored information. This approach helps keep the system modular and provides a structured method for handling application logic.

MySQL is used as the relational database management system for storing and managing application data. In PRAGATI AI, the database is used to maintain user information, authentication details, and chat-related records. It provides structured storage and allows the system to efficiently retrieve and manage required information. Database tables help maintain organized records and support secure access to stored information. The database is an important part of the system because it provides persistent storage for information required by the application.

XAMPP is used as the local development and testing environment for PRAGATI AI. It provides the required environment for running the PHP-based application and connecting the system with the MySQL database. During development, XAMPP allows the project to be executed locally so that different modules such as signup, login, database connectivity, and chatbot operations can be tested. It combines the required server-side components and provides a convenient environment for application development and testing.

The AI API is used to provide the intelligent chatbot functionality of PRAGATI AI. The chat interface sends user questions to the connected AI service through the backend, and the AI service processes the request and returns an appropriate response. This integration allows the system to provide real-time AI-powered assistance to users. The API-based architecture also separates the AI service from the main application interface and allows the backend to manage communication between the user and the AI engine.

The selected technologies work together through a client-server architecture. The frontend provides the interface through which users interact with the system, while the backend processes user requests and communicates with the database and AI service. HTML, CSS, and JavaScript provide the user interface and client-side interaction, whereas PHP manages backend operations. MySQL maintains user and chat information, and the AI API provides intelligent responses. This combination creates a structured environment for implementing the different functions of PRAGATI AI.

PRAGATI AI also includes additional interactive capabilities within the chat interface. The system supports multilingual communication in English, Hindi, and Gujarati, allowing users to ask questions and interact with the chatbot in their preferred language. It also supports file upload and interaction with uploaded content so that users can use their learning materials during the conversation. Microphone-based input allows users to speak their questions instead of typing them, while the voice-listening functionality provides an additional way of interacting with the chatbot.

The combination of these programming languages and technologies makes PRAGATI AI a practical and user-friendly web-based learning platform. Each technology performs a specific role in the development of the system, from creating the interface and handling user interaction to processing backend operations, storing information, and generating AI-based responses. The modular use of these technologies also allows the system to be maintained and expanded with additional features in the future. The overall design focuses on simplicity, security, interactivity, and scalability.

Table 3.3: Programming Languages/Frameworks

S. No.	Programming Language / Framework	Purpose / Usage
1	XAMPP	Used to structure the web pages and define the content and layout of the PRAGATI AI application interface
2	AI API	Used to connect the chatbot with an AI service for processing user queries and generating real-time responses.
3	**HTML**	Used to structure the web pages and define the content and layout of the application interface.
4	**CSS**	Used to style the application, including page layouts, forms, tables, buttons, and other visual elements.
5	**JavaScript**	Used to add client-side interactivity and dynamic behaviour to the web interface.
6	**SQL**	Used for creating queries and performing database operations such as inserting, retrieving, updating, and deleting business records in MySQL.
7	PHP	Used as the server-side scripting language for backend processing, user authentication, and handling application requests.

CHAPTER 4: SYSTEM ANALYSIS AND DESGIN

4.1 Problem Definition

PRAGATI AI is developed to address the limitations of traditional learning methods where students may have to depend on fixed classroom hours and direct teacher assistance for resolving academic doubts. The system provides an AI-powered chatbot through which students and teachers can ask questions and receive responses in real time. The proposed platform provides secure signup and login functionality to ensure controlled access to the system. The system connects the chat interface with an AI API to process user queries and provide relevant answers. It aims to reduce delays in doubt resolution and provide learning support beyond regular classroom interaction. PRAGATI AI also supports multilingual communication in English, Hindi, and Gujarati, allowing users to interact in their preferred language. Users can upload files and interact with the content of the uploaded files through the chatbot. The system also provides microphone-based input so that users can ask questions using their voice instead of typing. The system uses frontend, backend, database, and API technologies together to provide an integrated learning environment. It is designed to support students and teachers through continuous and convenient academic assistance. The proposed system focuses on improving accessibility, interaction, and learning convenience through AI technology. Overall, PRAGATI AI addresses the need for a more flexible, interactive, and accessible digital learning platform.

4.2 Functional Requirements and Non- Functional Requirements

The functional requirements define the main operations that PRAGATI AI must perform, including secure signup and login, user authentication, real-time chatbot interaction, AI API integration, database management, and session handling. The system should allow students and teachers to ask questions and receive AI-generated responses through the chat interface. It should also support multilingual interaction, file upload and content interaction, and voice-based input features. The non-functional requirements focus on performance, security, usability, reliability, and scalability. The system should provide quick responses, protect user credentials and API information, maintain stable operation, and provide a simple and responsive interface.

Table 4.2.1: Functional Requirements and Non-Functional Requirements

S. No.	Functional Requirement	Description
1	User Registration	The system should allow users to create an account by providing the required registration details.
2	User login	The system should allow registered users to log in securely using their credentials.
3	User Authentication	The system should verify user credentials and provide access only to authorized users.
4	Role Management	The system should identify users as students or teachers and provide appropriate access to the system.
5	AI Chatbot	The system should allow users to ask academic questions and receive AI-generated responses in real time.
6	Chat History	The system should store user queries and AI-generated responses for future reference.
7	Multilingual Interaction	The system should support user interaction in English, Hindi, and Gujarati.
8	File Upload and Interaction	The system should allow users to upload files and interact with the information contained in the uploaded content.
9	Voice Input	The system should allow users to ask questions using microphone-based voice input.
10	Voice Listening	The system should provide voice-listening functionality for more interactive communication with the chatbot.

Table 4.2.2: Functional Requirements and Non-Functional Requirements

S. No.	Non-Functional Requirement	Description
1	Usability	The system should provide a simple, clear, and easy-to-use interface for students and teachers.
2	Performance	The system should respond to user requests efficiently and provide chatbot responses within a reasonable time.
3	Reliability	The system should perform its intended functions consistently during normal usage.
4	Data Accuracy	The system should maintain accurate and consistent user and chat-related information.
5	Maintainability	The application should have a structured and modular design that makes maintenance and future modifications easier.
6	Scalability	The system should allow additional features and modules to be added in future versions.
7	Compatibility	The system should work properly across commonly used devices such as desktops, laptops, and tablets.
8	Security	The system should protect user credentials, sessions, stored data, and API-related information from unauthorized access.
9	Availability	The system should remain accessible to users whenever the required server, internet connection, and AI service are available.
10	Responsiveness	The user interface should provide smooth interaction and properly respond to user actions without unnecessary delays.

4.3 Feasibility Study

PRAGATI AI is feasible because it can be developed using commonly available technologies such as HTML, CSS, JavaScript, PHP, MySQL, and XAMPP. The system is technically feasible, easy to operate, and can be developed using basic computer hardware and a stable internet connection. The use of open-source tools also keeps the development cost low. The project can be completed within a reasonable time because its modules are structured and manageable.

Table 4.3: Feasibility Study

S. No.	Type of Feasibility	Description
1	**Technical Feasibility**	The project is technically feasible because it uses commonly available technologies such as HTML, CSS, JavaScript, PHP, and MySQL. The application can be developed and operated on a standard computer without specialized hardware.
2	**Economic Feasibility**	The project is economically feasible because the major technologies and development tools used are open-source and freely available. This helps reduce software licensing and development costs.
3	**Schedule Feasibility**	The project is feasible within the available development time because the system consists of manageable modules such as authentication, chatbot interaction, database management, and API integration.
4	**Operational Feasibility**	The system is operationally feasible because it provides a simple and user-friendly interface for students and teachers. Users can access authentication and chatbot features with minimal technical knowledge.

4.4 System Architecture

PRAGATI AI follows a client-server architecture in which the user interacts with the system through the frontend interface. The frontend is developed using HTML, CSS, and JavaScript to provide a simple and interactive user experience. User requests from the chat interface are sent to the backend for processing. PHP manages the backend operations and handles communication between the frontend, database, and AI service. MySQL is used to store user information, authentication details, and chat history. The backend sends chatbot queries to the connected AI API for processing. The AI service generates the required response and sends it back through the backend. The response is then displayed to the user through the chat interface in real time. The architecture also supports multilingual interaction, file upload and content interaction, and voice-based input through the chatbot interface.

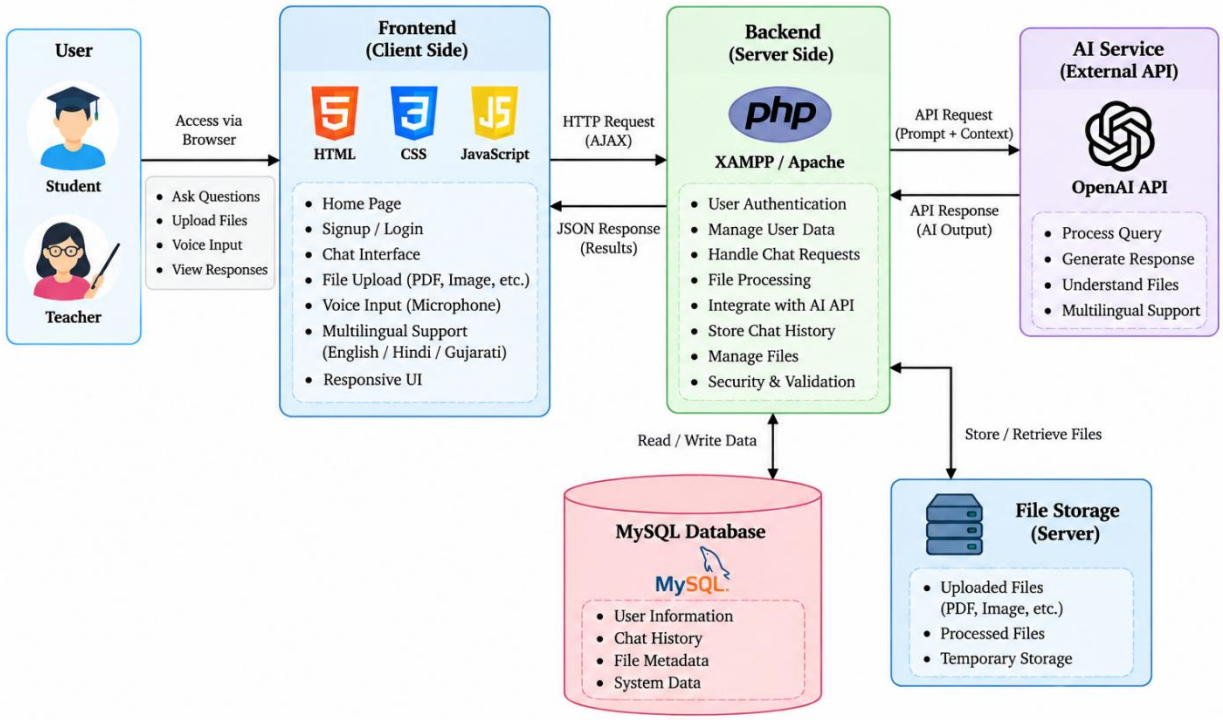


Fig 4.1: System Architecture

4.5 **UML Diagrams**

UML diagrams are used to represent the structure and interaction of different components of the PRAGATI AI system. The diagrams help describe how users interact with the system and how different modules work together. The main users of the system are students and teachers who can register, log in, and access the chatbot. The system includes modules such as user authentication, role management, chat interaction, AI integration, and database management. These diagrams provide a clear visual representation of the system and help in understanding its overall design and functionality.

4.5 ER Diagram

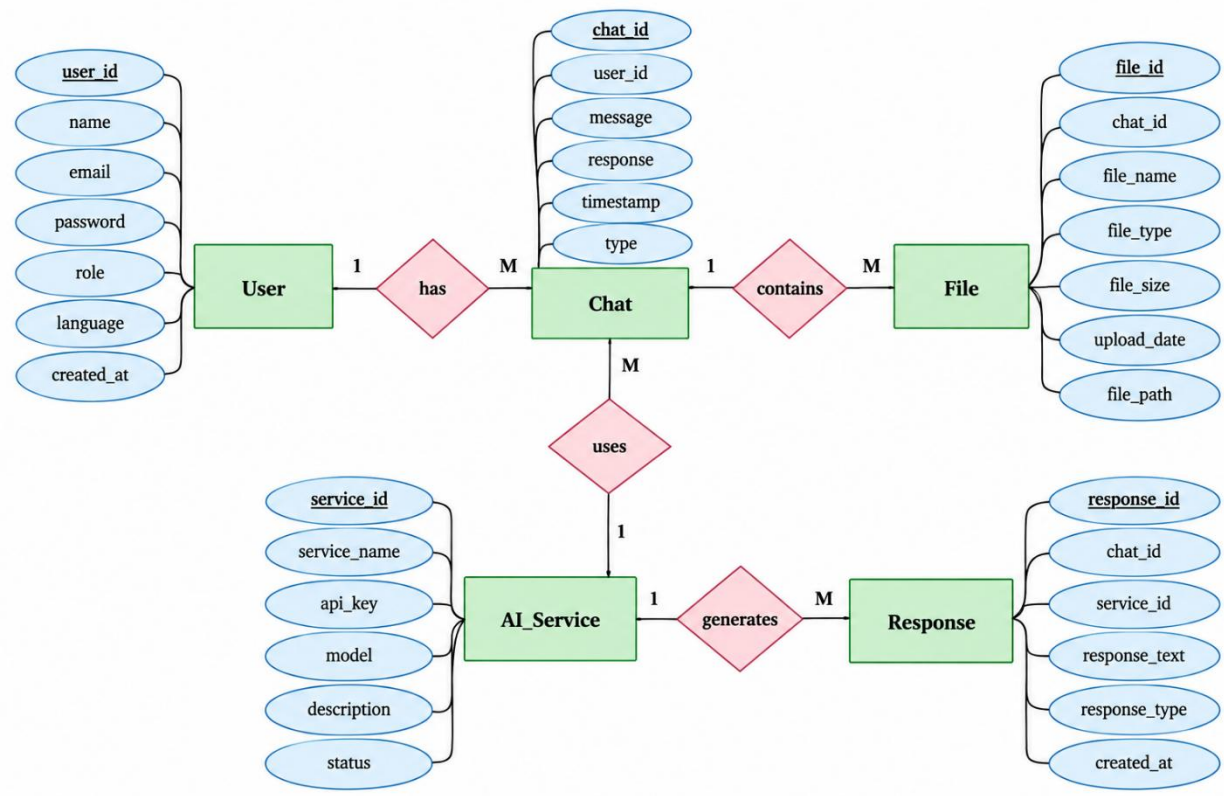


Fig 4.2: ER Diagram

4.6 Database Design

PRAGATI AI database is designed to store and manage user information, authentication details, and chat interactions with the AI system. MySQL is used as the relational database management system for storing and retrieving the required data. The Users table maintains information of students and teachers registered on the platform, including user ID, full name, password, role, and account creation date. The user ID acts as the primary key and uniquely identifies each registered user. The role field is used to identify whether the user is a student or a teacher. The Chat table contains fields such as chat ID, user ID, user message, AI response, and timestamp, allowing the system to maintain a record of interactions. The user ID connects chat records with the The corresponding registered user and helps organize the stored information. The database design is structured to provide secure, organized, and efficient storage for the core information used by PRAGATI AI. Overall, the database supports the functionality of the application and provides a foundation for future expansion of the system.

Table 4.4: Database Design

S. No.	**Entity / Table**	**Purpose**
1	Users	Stores information about students and teachers registered on the PRAGATI AI platform.
2	User Authentication	Maintains login credentials and supports secure user authentication and access control.
3	Chat	Stores user queries and AI-generated responses exchanged through the chatbot.
4	Chat History	Maintains previous chat interactions for future reference and retrieval
5	User Roles	Stores and manages the role of each user, such as student or teacher, for appropriate access to system features.

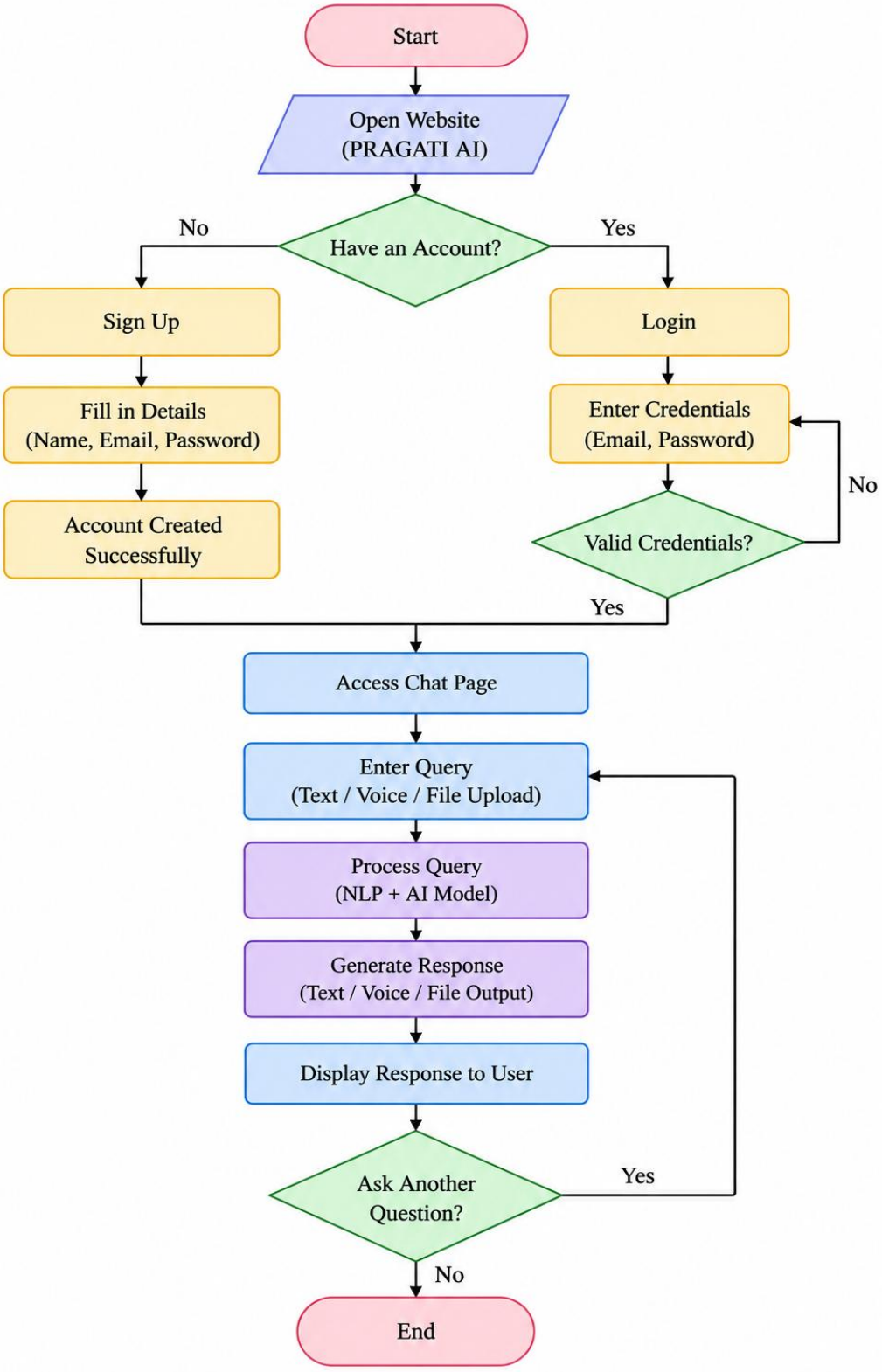
4.7 Flowcharts/ Algorithms

Flowcharts and algorithms are used to represent the logical flow of the Business Inventory & Sales Analytics System. They describe how the system processes user requests, manages inventory, handles orders, and updates sales information. The flow begins with the user interacting with the web interface, followed by request processing through the Flask backend and database operations in MySQL.

Table 4.5: Flowchart/Algorithms

Step	Process
1	User opens the PRAGATI AI web application.
2	User creates an account or logs in using valid credentials.
3	The user accesses the chatbot interface after successful authentication.
4	User enters a question through text, microphone input, or uploads a file for interaction
5	The request is sent to the backend for processing and validation.
6	The backend communicates with the AI service and retrieves the required response or processes the uploaded content.
7	The backend communicates with the AI service and retrieves the required response or processes the uploaded content.
8	The user can continue the conversation, use another language, upload additional content, or ask another question.

Fig 4.3: Flowchart



CHAPTER 5: IMPLEMENTATION

5.1 Modules Description

PRAGATI AI is divided into different modules to organize and manage the major functions of the system. The User Authentication Module handles user registration, login, credential verification, and session management to provide secure access to the platform. The Role Management Module identifies whether the user is a student or teacher and provides access according to the assigned role. The Chat Interaction Module provides the main chatbot interface through which users can ask questions and receive AI-generated responses. The AI Integration Module connects the chatbot with the AI service through an API and processes user queries to generate responses in real time. The Database Management Module uses MySQL to store and manage user information and chat-related data. The User Interface Module is developed using HTML, CSS, and JavaScript to provide a simple, responsive, and easy-to-use interface. The Session and Security Module manages user sessions and protects sensitive information from unauthorized access. PRAGATI AI also includes multilingual communication in English, Hindi, and Gujarati, file upload and interaction with uploaded content, microphone-based voice input, and voice-listening functionality within the chatbot interface. Together, these modules work in an integrated manner to provide a secure, interactive, and efficient AI-powered learning environment.

5.2 Software Implementation

The PRAGATI AI system is implemented using frontend, backend, database, and AI technologies. HTML is used to create the structure of the application pages, while CSS is used for styling, layout, and visual presentation. JavaScript provides interactive functionality and smooth user interaction. PHP is used for backend processing, authentication, session handling, and communication with the database. MySQL is used to store user information, authentication details, and chat-related data. XAMPP provides the local development environment for running and testing the application. The chatbot is connected to an AI service through an API for processing user queries and generating responses in real time. The implementation also includes secure signup and login, role-based access, and continuous chatbot interaction. PRAGATI AI further supports English, Hindi, and Gujarati communication, file upload and content interaction, microphone-based input, and voice-listening functionality within the chat interface. These technologies and features work together to provide a simple, secure, interactive, and efficient AI-powered learning platform.

5.3 Screenshots

The following screenshots demonstrate the working interface and major features of the PRAGATI AI web application The screenshots present the login and chatbot interface through which users can interact with the system and receive AI generated responses The application also provides file upload functionality that allows students to interact with their study materials In addition the system supports microphone input and multilingual conversation in English Hindi and Gujarati .

Figure 5.3.1: Home page

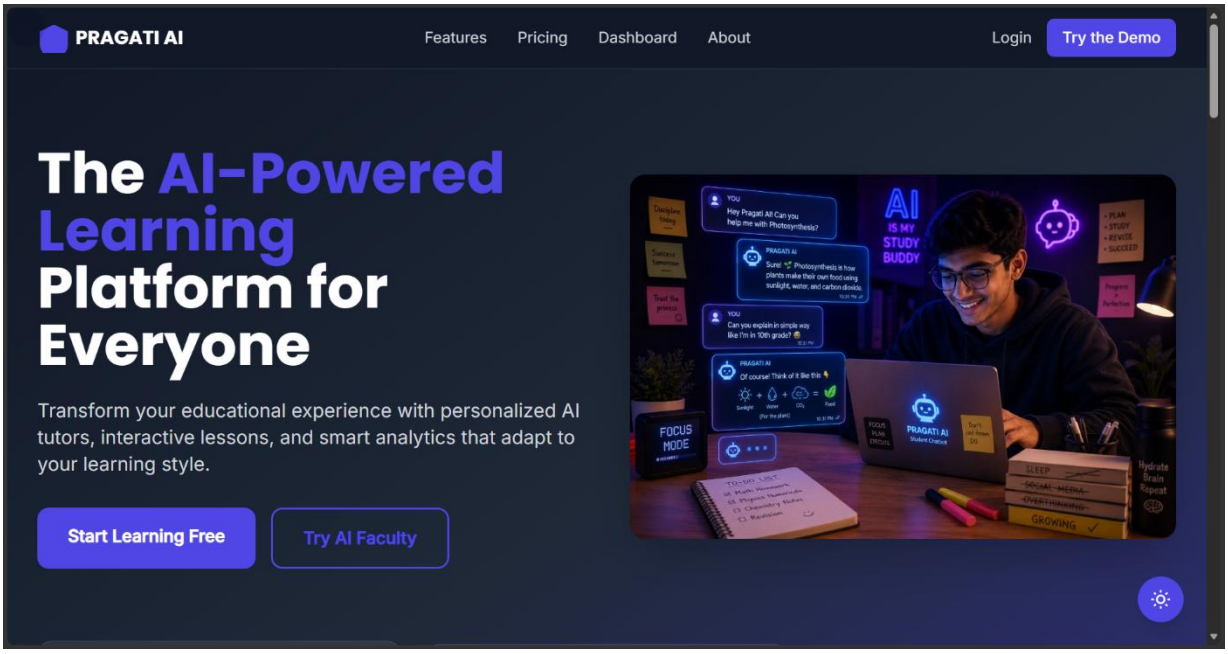


Figure 5.3.2: Chat page

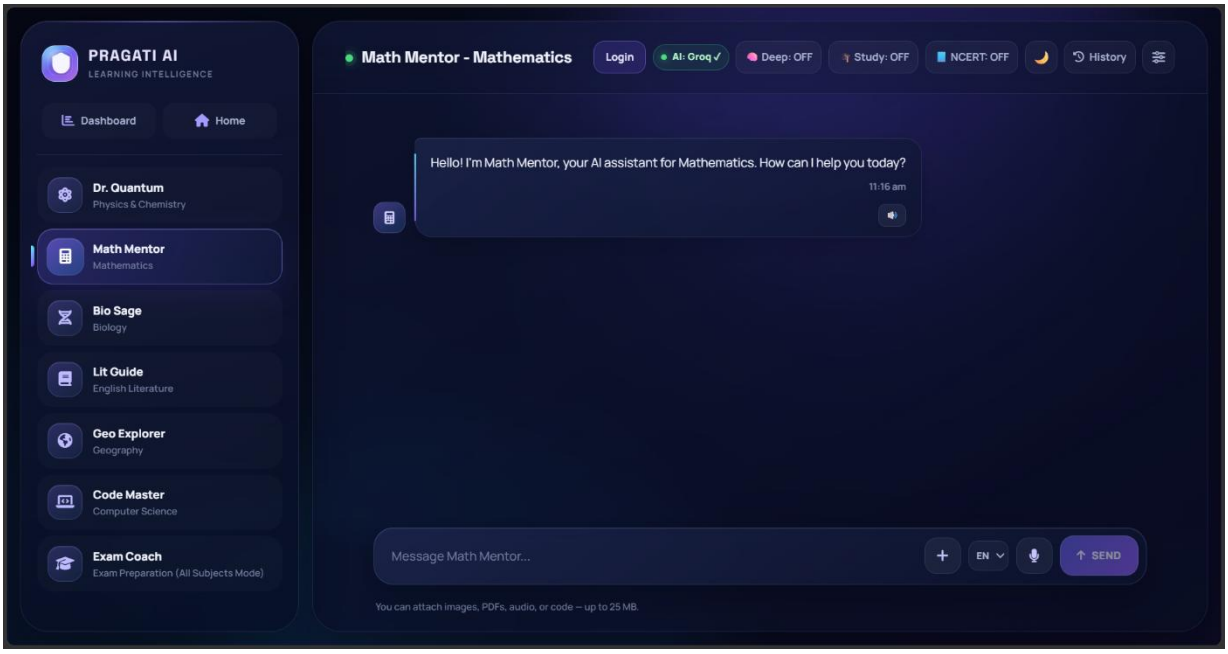


Figure 5.3.3:Login page

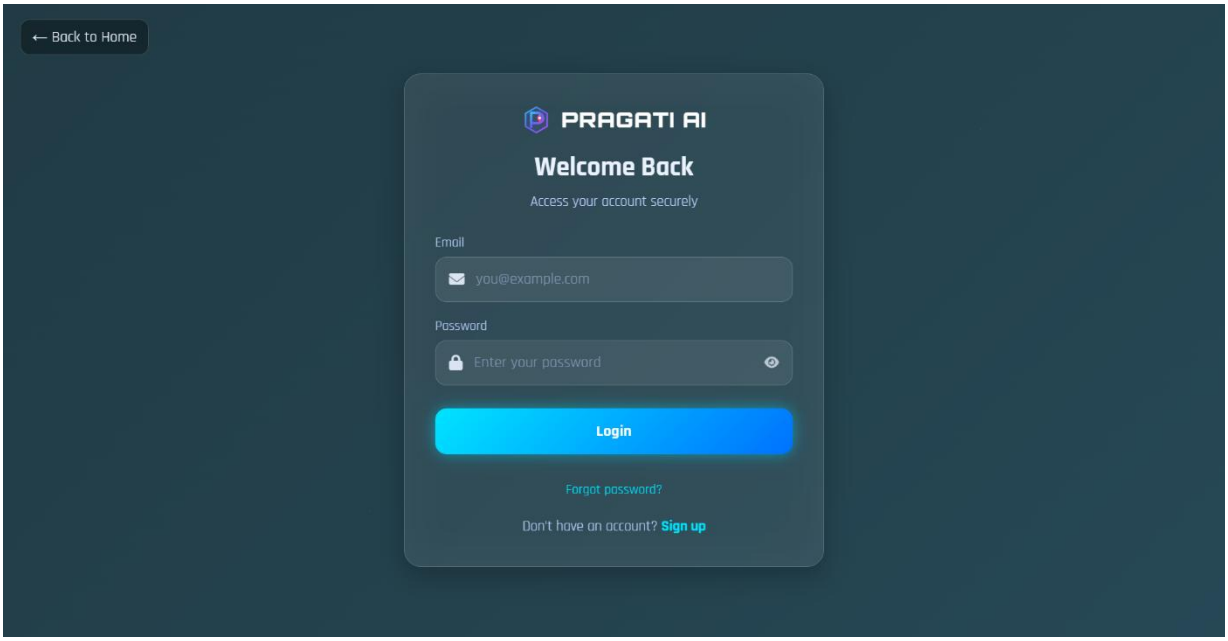
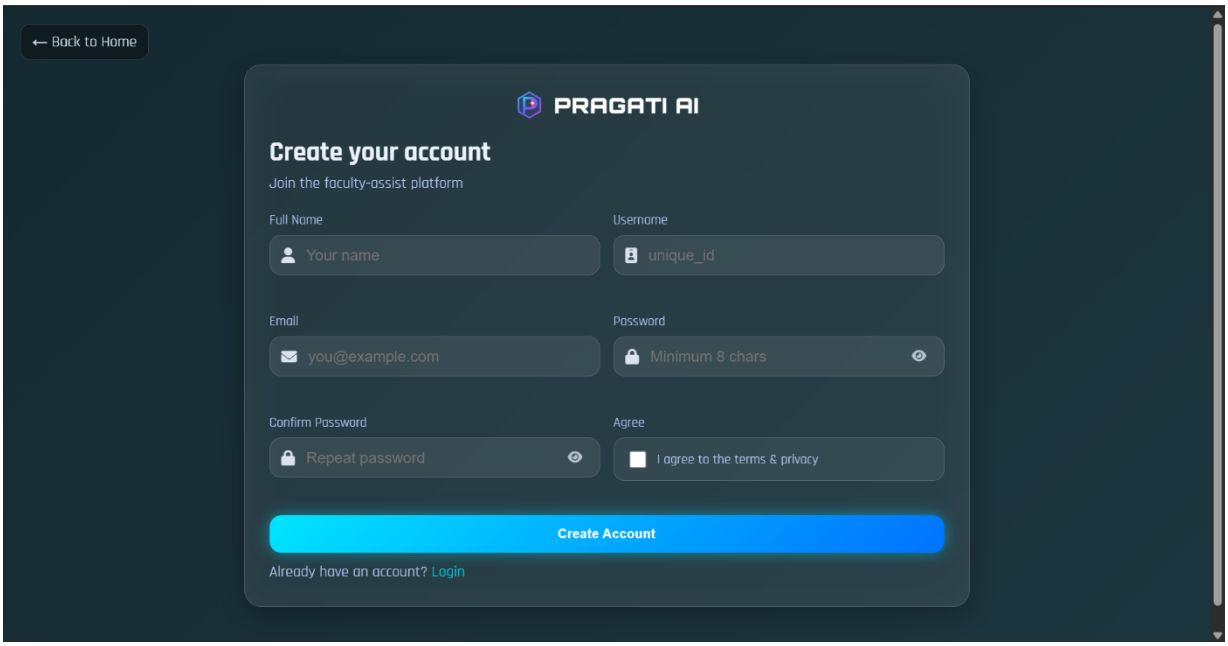


Figure 5.3.3: Sign up page



CHAPTER 6: RESULTS AND DISCUSSION

6.1 RESULTS

The implementation of PRAGATI AI resulted in a functional AI-powered learning platform designed to support students and teachers through an interactive chatbot. The system provides secure signup and login functionality through which registered users can access the platform. After successful authentication, users can access the main chat interface and submit academic questions. The chatbot is connected to an AI service through an API, allowing user queries to be processed and responses to be generated in real time. This provides users with quick academic assistance and reduces dependence on external resources during learning sessions.

The implemented system provides a simple and user-friendly interface that supports smooth navigation between the major pages of the application. The frontend is developed using HTML, CSS, and JavaScript, while PHP manages backend operations and MySQL stores user and chat-related information. The system follows a modular structure in which authentication, chat interaction, database management, and AI integration work together as separate but connected components. This structure makes the application easier to manage and provides a foundation for future expansion.

One of the main results of the system is the availability of real-time AI-based academic assistance. Users can enter questions through the chat interface and receive AI-generated responses without requiring direct manual assistance from a teacher for every query. The system is therefore useful for self-paced learning as well as classroom support. The original project design also identifies the chatbot as the main interface for asking academic questions and receiving real-time solutions through API integration.

PRAGATI AI also provides the additional implemented interaction features of multilingual communication in English, Hindi, and Gujarati. Users can interact with the chatbot according to their preferred language, making the system more convenient for different users. The platform also supports file upload and interaction with uploaded content, allowing users to use their own learning materials during the conversation. Microphone-based input provides another method of asking questions, while the voice-listening functionality makes the chatbot interaction more flexible and interactive.

The system successfully maintains user and chat-related information through the MySQL database. User authentication details and chat interactions can be stored and retrieved when required. The database structure includes user information and chat records, which supports organized management of the information generated through the platform. This contributes to secure access, efficient data handling, and future expansion of the system.

The developed system is also practical from a technical and operational point of view. It uses widely available technologies and can run in a common development environment such as XAMPP. The project does not require specialized hardware, and the use of commonly available software tools makes development and testing easier. The original project analysis also identifies the system as technically, operationally, economically, and schedule feasible.

6.2 DISCUSSION

The results show that PRAGATI AI achieves its primary objective of providing an accessible and interactive learning assistant through AI technology. The secure authentication system provides controlled access, while the chatbot offers immediate responses to user queries. The combination of a simple interface, database management, and AI integration creates a centralized learning environment for students and teachers. The system therefore addresses some of the limitations of traditional learning methods, particularly the delay in resolving academic doubts and the dependence on fixed classroom interaction.

The interactive nature of the system can also improve user engagement because users can continue asking questions through the same chatbot interface. The original system was designed to provide a simple learning environment that supports both teaching and self-paced learning, while reducing dependence on external resources during learning sessions. The addition of multilingual, file-based, and voice-based interaction in PRAGATI AI further increases flexibility by providing multiple ways for users to communicate with the assistant.

However, the system remains dependent on the availability of the AI API and a stable internet connection. If the API is slow or temporarily unavailable, responses may be delayed or may not be generated.

Security is another important aspect of the discussion. Since the system manages user credentials, sessions, chat information, and API communication, proper security measures are required to protect sensitive information. The source project recommends strong password validation, secure password storage, session management, and secure handling of the API key. These measures are important for maintaining user trust and preventing unauthorized access to the system.

The system also has opportunities for further improvement. The original analysis identifies the absence of advanced analytics, progress tracking, personalized dashboards, and administrative controls as areas that can be developed in future versions. Additional improvements can include stronger authentication, real-time analytics, quiz modules, progress dashboards, and an administration panel.

Overall, the results and discussion indicate that PRAGATI AI provides a practical and interactive foundation for AI-assisted learning. The system successfully combines secure authentication, chatbot interaction, database management, AI integration, multilingual communication, file-based interaction, and voice-based input in one learning platform. Its modular design allows the existing system to be maintained and expanded with additional educational features. The project therefore demonstrates the practical use of AI technology to make digital learning more accessible, interactive, and convenient for students and teachers.

CHAPTER 7: CONCLUSION

PRAGATI AI is an AI-powered learning platform developed to provide students and teachers with a simple, interactive, and accessible environment for academic assistance. The system provides secure signup and login functionality, allowing authorized users to access the platform. After login, users can interact with the AI chatbot through the chat interface and receive AI-generated responses in real time. The system uses HTML, CSS, JavaScript, PHP, MySQL, and an AI API to support the frontend, backend, database, and chatbot functionality.

PRAGATI AI supports multilingual communication in English, Hindi, and Gujarati, allowing users to interact with the chatbot in their preferred language. The system also provides file upload and interaction with uploaded content, microphone-based question input, and voice-listening functionality through the chat interface. These features provide multiple methods of interaction and make the learning process more flexible and convenient. The system follows a modular structure in which authentication, chat interaction, database management, and AI integration work together to provide an organized learning environment.

Overall, PRAGATI AI provides a practical and user-friendly solution for AI-assisted learning. It helps users receive quick academic assistance, supports self-paced learning, and provides continuous interaction through the chatbot. The combination of secure authentication, AI-based responses, database management, multilingual communication, file interaction, and voice-based features makes PRAGATI AI an effective and interactive learning platform.

CHAPTER 8: FUTURE SCOPE

The future scope of PRAGATI AI includes several improvements that can make the platform more powerful, secure, interactive, and useful for students and teachers. Advanced analytics can be integrated to track user activity, learning patterns, and academic performance. Progress dashboards can be added to help students and teachers monitor learning progress more clearly. Quiz and assessment modules can be introduced to make learning more interactive and measurable. An administrative panel can be developed for managing users, monitoring system activity, maintaining database information, and handling technical issues. Security can be strengthened through password recovery, email verification, and multi-factor authentication. A more advanced voice assistant can be developed so that users can directly talk with the chatbot through a natural two-way voice conversation instead of mainly depending on text input. The voice assistant can understand spoken questions and provide spoken responses, creating a more natural and convenient interaction. The system can also be improved to maintain conversation context for longer and more detailed interactions. Integration with learning management systems and other educational platforms can provide a connected digital learning environment. Additional educational tools and resources can also be integrated to support different learning requirements. The platform can be enhanced to support a larger number of users while maintaining reliable performance and secure data management. These developments can improve the accessibility, functionality, scalability, and overall effectiveness of PRAGATI AI.

Reference

- W3Schools – HTML, CSS, and JavaScript documentation for front-end development.
- PHP Official Documentation – Guidelines for server-side scripting, form handling, and API integration.
- MySQL Documentation –Database design, data storage, and secure access management.
- XAMPP by Apache Friends –Local development server setup and testing environment.
- OpenAI API Documentation / Chatbot API Docs –Integration of intelligent AI responses through API keys.
- MDN Web Docs – Web standards, JavaScript event handling, and UI design Principles.
- Software Engineering Principles – System design, testing, and deployment practices used for structured development.

Appendix

1. TECHNOLOGY USED

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	PHP, MySQL
AI Integration	Groq API Key
Version Control	Git

2. USER ROLES AND FUNCTIONABILITIES

Role	Functionalities
Student	Signup/Login to access the platform Ask academic questions via chat View AI-generated responses in real time Track previous chat history Update personal profile
Teacher	Signup/Login to access the platform Ask questions and verify responses Monitor students' queries and interactions Access class-specific data Provide feedback on AI responses
Administrator	Manage user accounts (add/remove) Monitor chat interactions and activity logs Maintain database integrity Configure and update API key settings Handle technical issues